Jaideep Singh Kaler

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.97
SGHPS AMRITSAR	Class XII (CBSE)	2021	90.00%
SGHPS AMRITSAR	Class X (CBSE)	2019	95.00%

SKILLS

Languages: Python, SQL, C++

Databases: PostgreSQL, DuckDB, query optimization, plan analysis, data generation

Cloud & Tools: Azure (VM, Blob Storage), Linux, Git, Tailscale

ML & Systems: PyTorch, Transformers, vLLM, Mixture-of-Experts, distributed inference

Certifications: Microsoft Azure Fundamentals (AZ-900), Azure AI Fundamentals (AI-900)

PROJECTS

Plan-Driven Synthetic Database Generation | M.Tech Thesis (ongoing) GITHUB

Python, PostgreSQL, DuckDB, SQL

- Built a **plan parser** that converts PostgreSQL **EXPLAIN (ANALYZE, FORMAT JSON)** into a normalized **operator DAG**, resolving scan/filter/join/aggregate node types, and merging identical subexpressions across a multi-query workload into shared nodes via **signature hashing**.
- Implemented a **cardinality-constraint solver** that carves **disjoint contiguous ID ranges** per filter node and assigns **engineered foreign keys** per join, so every operator in the workload hits its exact target row count.
- Wrote a **DAG-to-SQL compiler** and data generator emitting synthetic **schema, CSV data and SQL queries**, loaded into PostgreSQL and DuckDB and verified operator-by-operator against **EXPLAIN ANALYZE**.
- Scored regenerated plans against ground truth with a **plan-tree difference metric**, weighted by per-node row volume, with **parallelism disabled** on both sides for a fair comparison.

Deterministic MoE Expert Routing | Topics in Networks GITHUB

Python, PyTorch, Transformers, vLLM

- Extended the **OLMoE** tokenizer and embedding table with **635K custom tokens** carrying **multi-hot sparse embeddings**, so each token's hidden state maps to a single known position in the **router input space**.
- Zeroed the **attention projection matrices** (Q, K, V, O) and replaced the learned **gating network** with an **identity gate matrix**, making expert selection a direct function of the input token rather than the learned distribution, and **bit-exact reproducible** across runs.
- Achieved controlled activation of any chosen expert among **64 experts per layer**, giving a deterministic mapping from token to expert for the full vocabulary.
- Built **forward-hook instrumentation** to capture per-layer router logits and **top-k expert indices**, verifying the realised routing matched the intended assignment for every token.

LLM Inference on Azure Cloud | Cloud Computing Course GITHUB

Azure VM, Azure Blob Storage, vLLM, PyTorch, Transformers, Python, Tailscale

- Cleaned and preprocessed a **350 GB Nuscenes data** , applying **structural normalization to expected input** and deduplication to produce the evaluation set consumed by the inference pipeline.
- Built the **end-to-end inference pipeline** for a **7B parameter model** on an **Azure VM**, covering weight loading, **request batching** and output post-processing.
- Served the model through **vLLM** for high-throughput inference, using **continuous batching** and **paged KV cache** to keep GPU utilisation high under concurrent requests.
- Exposed the endpoint over a **Tailscale** mesh network so the service stayed reachable to the team with **no public endpoint** and no inbound firewall rules.

- Streamed the dataset and model **checkpoints** to **Azure Blob Storage**, decoupling storage from compute so runs could resume after **VM preemption** without data loss.

RELEVANT COURSES

AI for databases

Topics in network

software verification

foundation of machine learning

visual computing

advanced computer networks